K4H818-update

A firmware update tool for the K4H818 radio module.

Overview

K4H818-update is a Python 3 program used to flash new firmware onto a K4H818 radio module. It fetches the firmware file directly from the K4H818_firmware GitHub repository, then flashes it using a program called avrdude. It auto-detects which hardware variant is connected (SHARI RevK PiX or SHARI RevK PiHat) and handles the correct reset sequence for that hardware before flashing.

Unlike K4H818-prog, K4H818-update does not print its own version number when it runs.

Location

The location of the K4H818-update program differs between the HamVOIP and ASL3 images:

- HamVOIP: Located in **/usr/local/sbin**
- ASL3: Located in **/home/<username>**

Running K4H818-update

K4H818-update always fetches a firmware file from GitHub before flashing -- it never flashes a file left over from an earlier session unless that file is the specific cached copy described below.

Usage	Effect
(none)	Always fetches the latest published firmware (K4H818_fw_latest.hex) fresh from GitHub, then flashes it. Never cached, so this always reflects whatever is currently published -- running it twice in a row genuinely re-downloads both times.
-<version> (e.g. -1.5.13)	Fetches a specific, named firmware version. If a copy was already downloaded on a previous run, the cached copy is used directly and no network access is needed. If not, it is downloaded once and cached in K4H818-update's own folder for future use.

Firmware Caching

These two usage modes exist because a published version and "whatever's newest" have different, incompatible needs. A specific version number, once published, never changes -- so caching it permanently is always safe, and avoids re-downloading the same bytes on every run. "Latest" is a moving target by definition -- caching it would risk running an increasingly stale copy forever after the first download, so it is deliberately never cached and always re-fetched.

Cached files are named K4H818_fw_<version>.hex and live in the same folder as K4H818-update itself (matching the HamVOIP/ASL3 locations above). They can be deleted safely at any time -- K4H818-update will simply re-download that version the next time it is requested.

Board Detection

K4H818-update checks for a SHARI RevK PiX first (a USB-serial adapter appearing as /dev/ttyUSB0 or similar) before checking for a SHARI RevK PiHat (/dev/ttyAMA0 or /dev/ttyS0), matching K4H818-prog's own detection order. This order matters: the PiHat's UART can be permanently present at the OS level even when no PiHat is actually wired to it, so checking for it first would wrongly misidentify a PiX system.

For a SHAR RevK PiHat specifically, K4H818-update also performs a GPIO-based hardware reset immediately before flashing, using whichever GPIO tool is actually available on the system (the 'gpio' command on HamVOIP, or the libgpiod Python module on ASL3). A RevK PiX needs no equivalent step -- avrdude's own connection to the USB-serial adapter handles the reset.

Requirements

Normally avrdude is automatically installed during the setup of a HamVOIP or ASL3 node using the K4H818 since avrdude must be installed before K4H818-update can flash anything. If it is missing, K4H818-update reports this directly and exits without attempting to flash:

To manually install avrdude use the following command:

sudo apt-get install avrdude (for ASL3)

or

pacman -S avrdude (for HamVOIP)

Error Messages

Message	Cause
avrdude is not installed.	See Requirements above.
no K4H818 found -- neither /dev/ttyAMA0 nor /dev/ttyUSB* is present.	No K4H818 hardware is connected, or the OS hasn't recognized it yet -- check the physical connection.
<url> returned HTTP 404.	The requested version number doesn't exist in the K4H818_firmware repository -- check for a typo, or that this version was actually published.
could not reach <url> (<reason>).	No network connectivity, or the K4H818_firmware repository is temporarily unreachable.
avrdude reported an error -- firmware update failed.	avrdude itself failed partway through flashing -- see the avrdude output printed just above this message for the specific cause.